

WASP-43b

R-1297

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_190404 · created 2026-07-31T16:51:18+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458577.76659 +/- 0.00049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1779 +/- 0.0071
Transit depth (Rp/Rs)^2	3.16 +/- 0.25 [%]
Orbital Inclination (inc)	87.07 +/- 0.69
Airmass coefficient 1 (a1)	1.015 +/- 0.011
Airmass coefficient 2 (a2)	-0.0096 +/- 0.0061
Scatter in the residuals of the lightcurve fit is	1.1 %
Best Comparison Star	#3 - [278, 247]
Optimal Aperture	4.9
Optimal Annulus	16.93
Transit Duration (day)	0.06156 +/- 0.00081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1779 ± 0.0071 vs 0.1594 (deviation 2.28σ)
Duration fit vs published	0.06156 d vs 0.0512 d (deviation 11.14σ)
Mid-time O–C	-4.2 min
Depth SNR	21.9
Residual scatter	1.1 %

Light curve

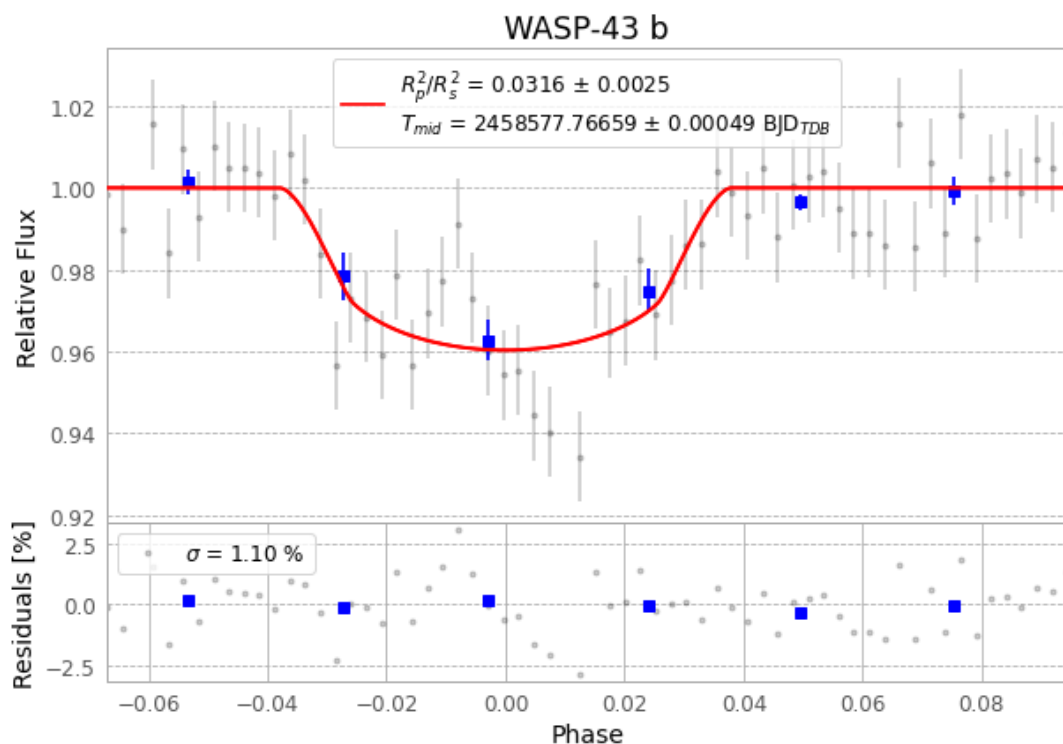


Figure 1 — FinalLightCurve_WASP-43 b_2019-04-03.png (SHA-256 18d7c4947752c8b0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [410, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ca595f0112811d7b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-43190404045712.FITS → WASP-43190404080612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-190404.log (75e24883a9b5...), FinalLightCurve_WASP-43 b_2019-04-03.png (18d7c4947752...), FinalLightCurve_WASP-43 b_2019-04-03.csv (8203a0273b80...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 16:51Z.